

or enemy-occupied territory than at any previous time in military history.

Yet another advantage of laser-guided munitions is their resistance to jamming and electronic counter-measures. Ability to operate at night, simplicity, reliability and maintainability are other positive attributes.

While laser-guided munitions are highly accurate and have demonstrated their efficacy in several conflicts in the past, the precision of attack and reliability of operation are guaranteed only under certain conditions. Factors that have a large bearing on the overall performance of a laser-guided munitions delivery mission include accurate and uninterrupted target designation, atmospheric attenuation due to prevailing environmental conditions and mode of weapon release.

Laser designation of the target should be such that the weapon's seeker head finds itself within the reflected radiation basket and also the received laser power is greater than its sensitivity. This is the first major challenge. Laser designator to target path length is sometimes an issue under adverse environmental conditions. Laser under these conditions is attenuated more than it would have under ideal or normal conditions. This may lead to laser power as received by the seeker falling below its threshold, which further causes failure of the guidance system if the received power is inadequate.

Improper designation may also lead to mishits. This was observed during the Gulf war when the laser radiation was reflected off sand surface rather than the intended target. Temporary blockage of laser radiation due to smoke, fog and dusty conditions and increased moisture content is another reason for guided munitions missing targets.

Correct weapon release is another challenge. The guidance system of earlier laser-guided munitions, such as Paveway-II, resulted in a rectilinear flight path that had a tendency to lag below the sight line. Currently, weapons are released on an unguided



Fig. 9: Diving top attack in the case of Krasnopol

ballistic flight path. The weapon release should be such as to allow the weapon on a ballistic trajectory find itself within the laser basket at the time of start of terminal guidance. Relatively narrow field-of-view of the laser seeker further complicates the problem.

Uninterrupted target designation is yet another challenge. The concept of terminal guidance reduces the overall designation time and improves reliability of uninterrupted designation. In the case of autonomous laser designation, that is, designation from weapon-carrying aircraft, uninterrupted target designation is a big problem in the presence of smoke, fog, clouds and dusty conditions. In addition, it leaves the aircraft vulnerable to enemy attack by ground fire or air support.

Optimum weapon release height poses another challenge. Optimum altitude for weapon release is in the range of 6km to 9km. This makes attack aircraft seriously vulnerable to surface-to-air missile (SAM) attack. During their 1981 raid on the Iraqi nuclear reactor at Osirak, the Israeli Air Force chose to use unguided Mark 84 bombs rather than laser-guided weapons during a raid because they felt that the need to designate the target would leave attackers unacceptably vulnerable.

Major laser-guided weapon systems

Laser-guided munitions are broadly categorised as surface-launched

projectiles, aerially-delivered bombs and surface-to-surface, surface-to-air and air-to-surface missiles. A large number of laser-guided weapon systems in these categories are manufactured by international giants, including Lockheed Martin, SAAB Bofors Dynamics Israel Aerospace Industries, Matra, Raytheon and KBP Instrument Design Bureau. The more common and established weapons, including Krasnopol/Krasnopol-M and Copperhead (cannon-launched laser-guided projectiles), Paveway-series and Griffin (aerially-delivered bombs), LAHAT, RBS-70/RBS-70NG and Hellfire (laser-guided missiles) are briefly discussed in the following paragraphs.

Laser-guided projectiles

Krasnopol and Copperhead are cannon-launched fin-stabilised terminally laser-guided explosive projectiles designed to engage small, hard, point-ground targets, such as tanks, armoured vehicles, self-propelled artillery systems and other high-value targets, such as bridges, defensive fortifications, C⁴ (command, control, communications, computers and intelligence) centres, etc.

Krasnopol projectile is produced in two variants, namely, Krasnopol and Krasnopol-M. The former is a 152mm two-section projectile designed to operate with both towed and self-propelled guns and howitzers. It, however, has a shortcoming that it is incompatible with 2S19 auto-loader due to the projectile's length.

Krasnopol-M is a 152mm/155mm projectile. It is an improvement over Krasnopol and is fully compatible with 2S19 auto-loader, which makes it usable with western produced 155mm howitzers.

Other than that, both have the same attack profile (diving top attack as illustrated in Fig. 9), targets engaged and the type of warhead used. The target ranges are similar; 20km in the case of Krasnopol and 17km for Krasnopol-M. Fig. 10 shows the